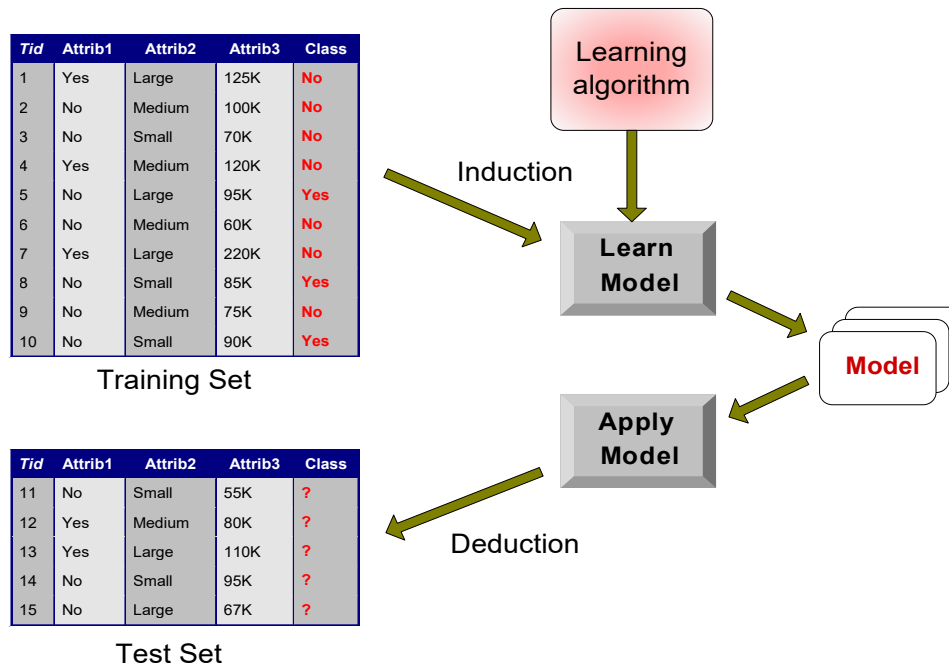
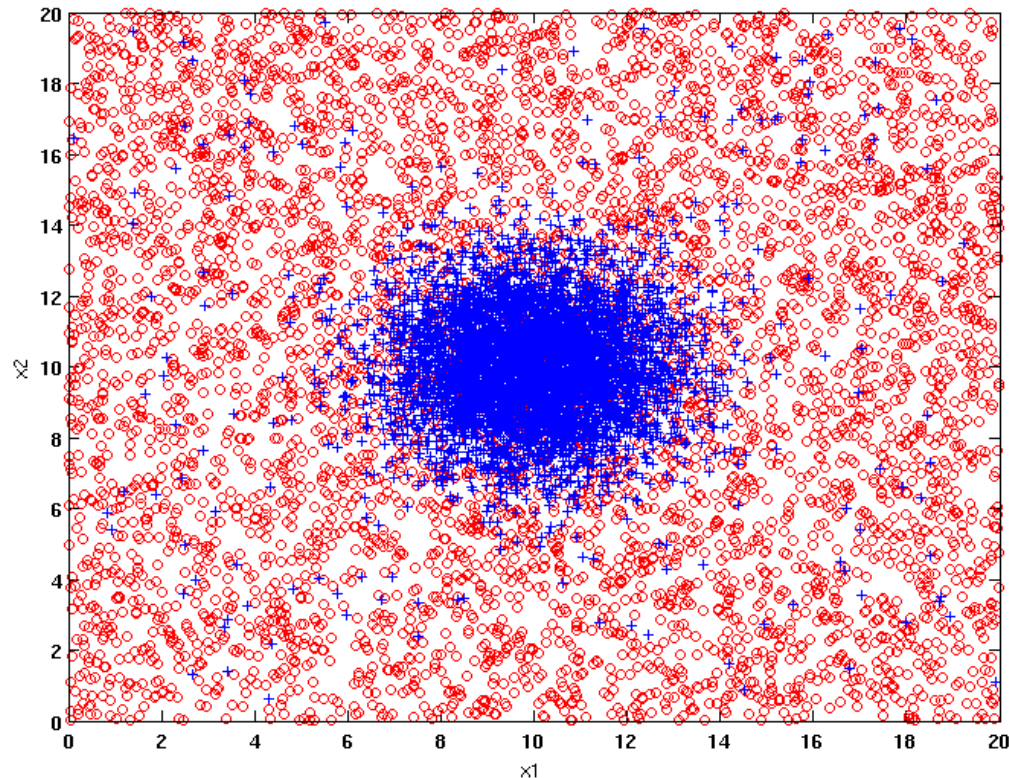

Model Overfitting

Classification Errors

- **Training errors:** Errors committed on the training set
- **Test errors:** Errors committed on the test set
- **Generalization errors:** Expected error of a model over random selection of records from same distribution



Example Data Set



Two class problem:

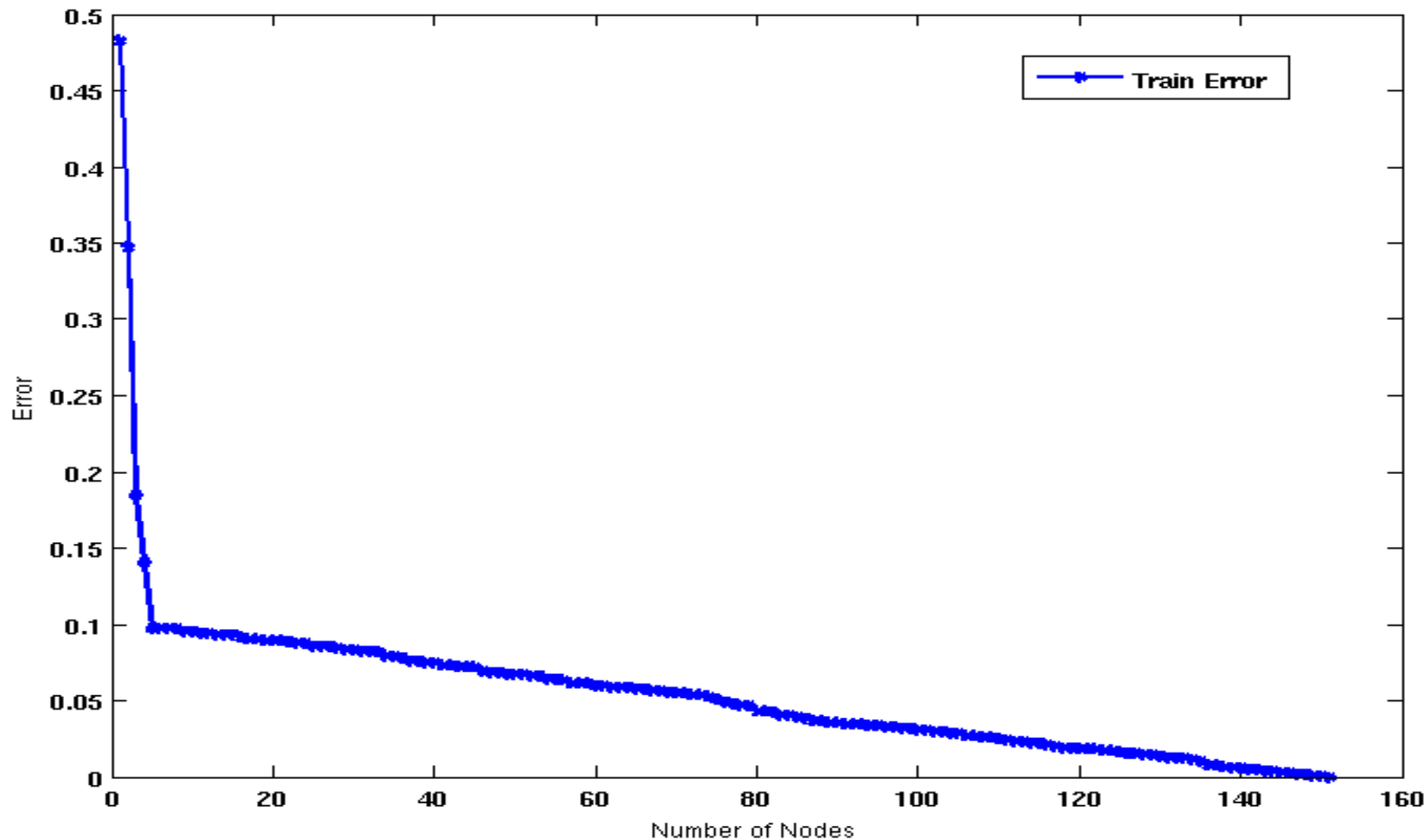
+ : 5400 instances

- 5000 instances generated from a Gaussian centered at (10,10)
- 400 noisy instances added

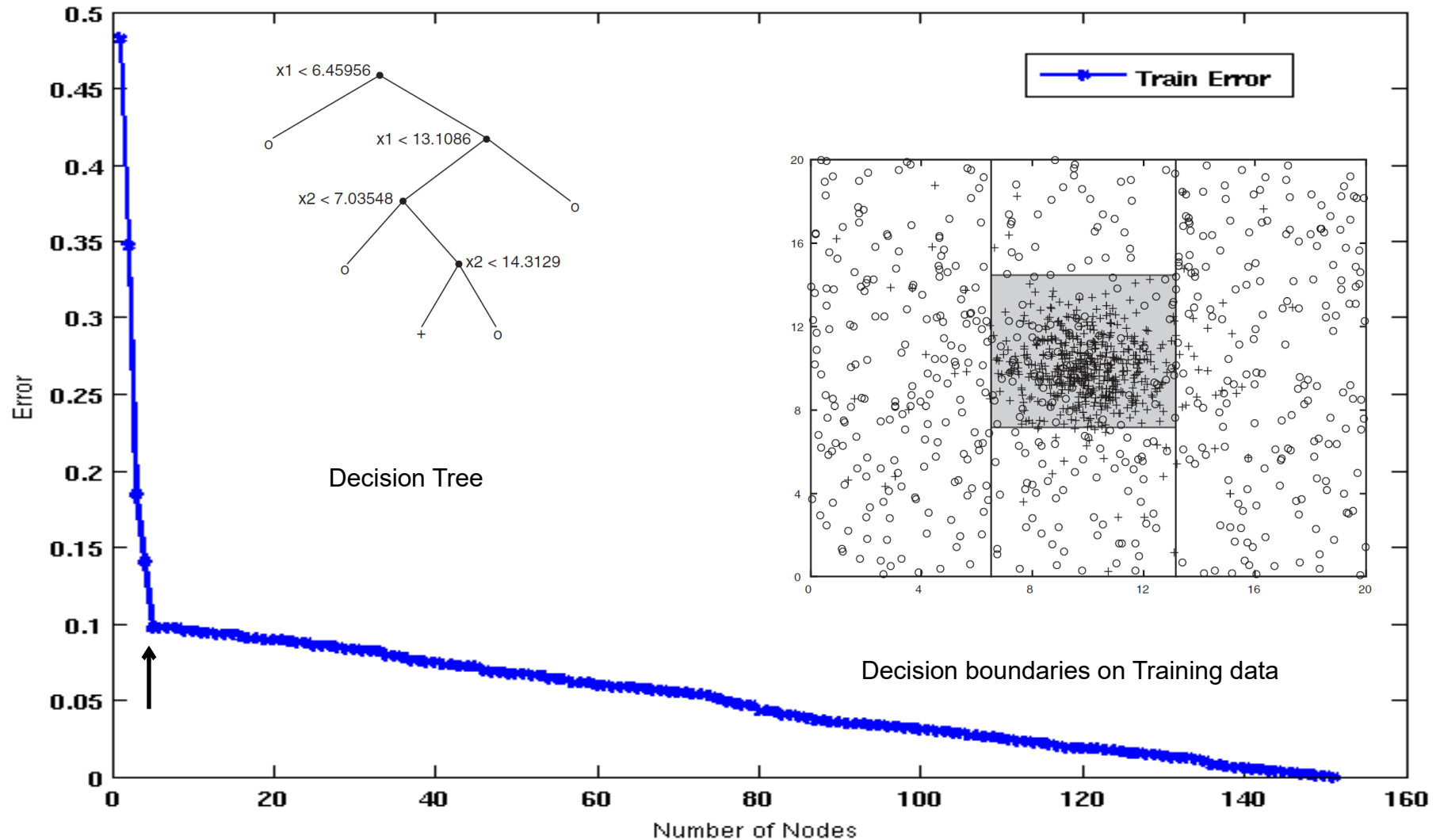
o : 5400 instances

- Generated from a uniform distribution

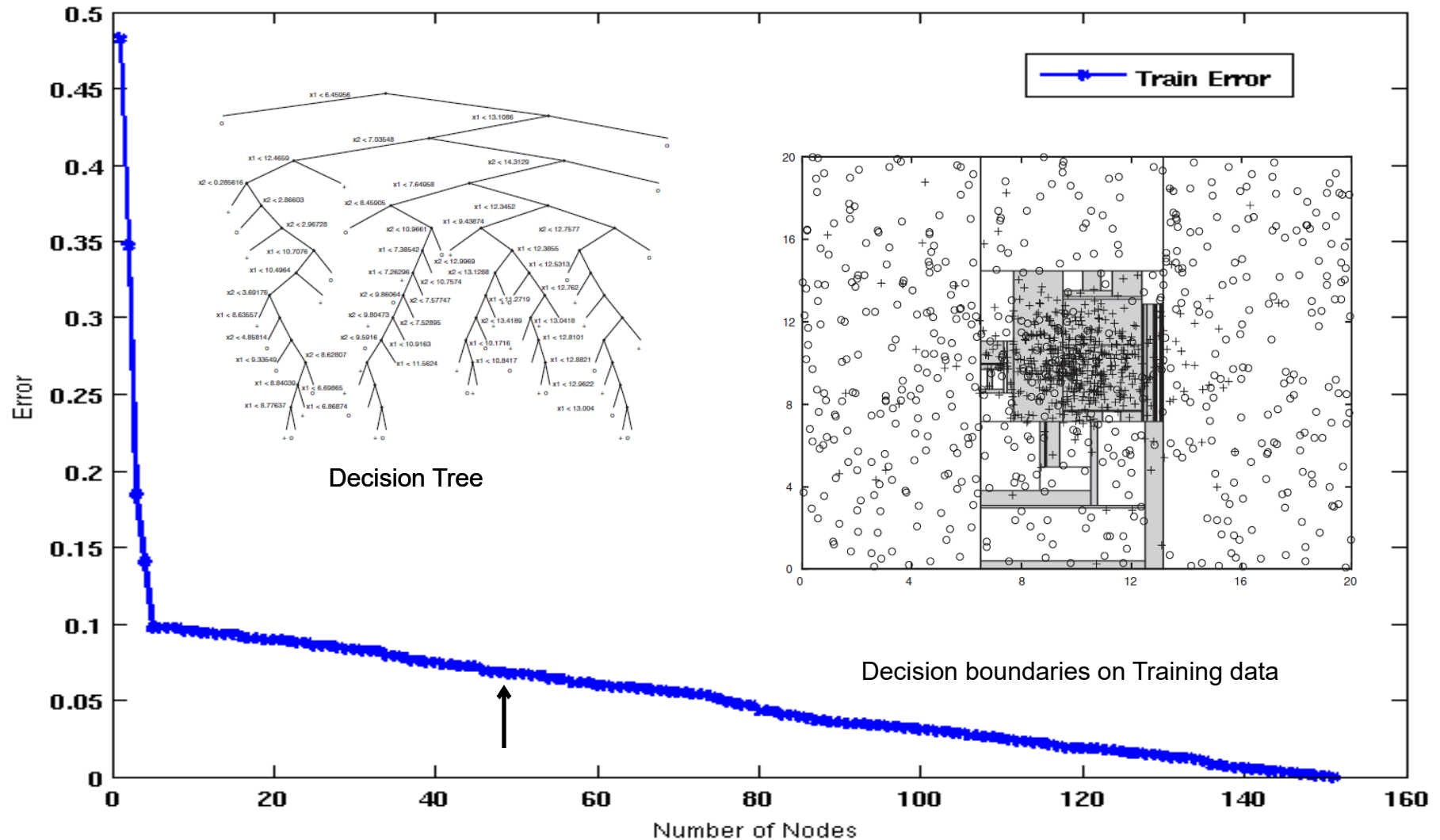
Increasing number of nodes in Decision Trees



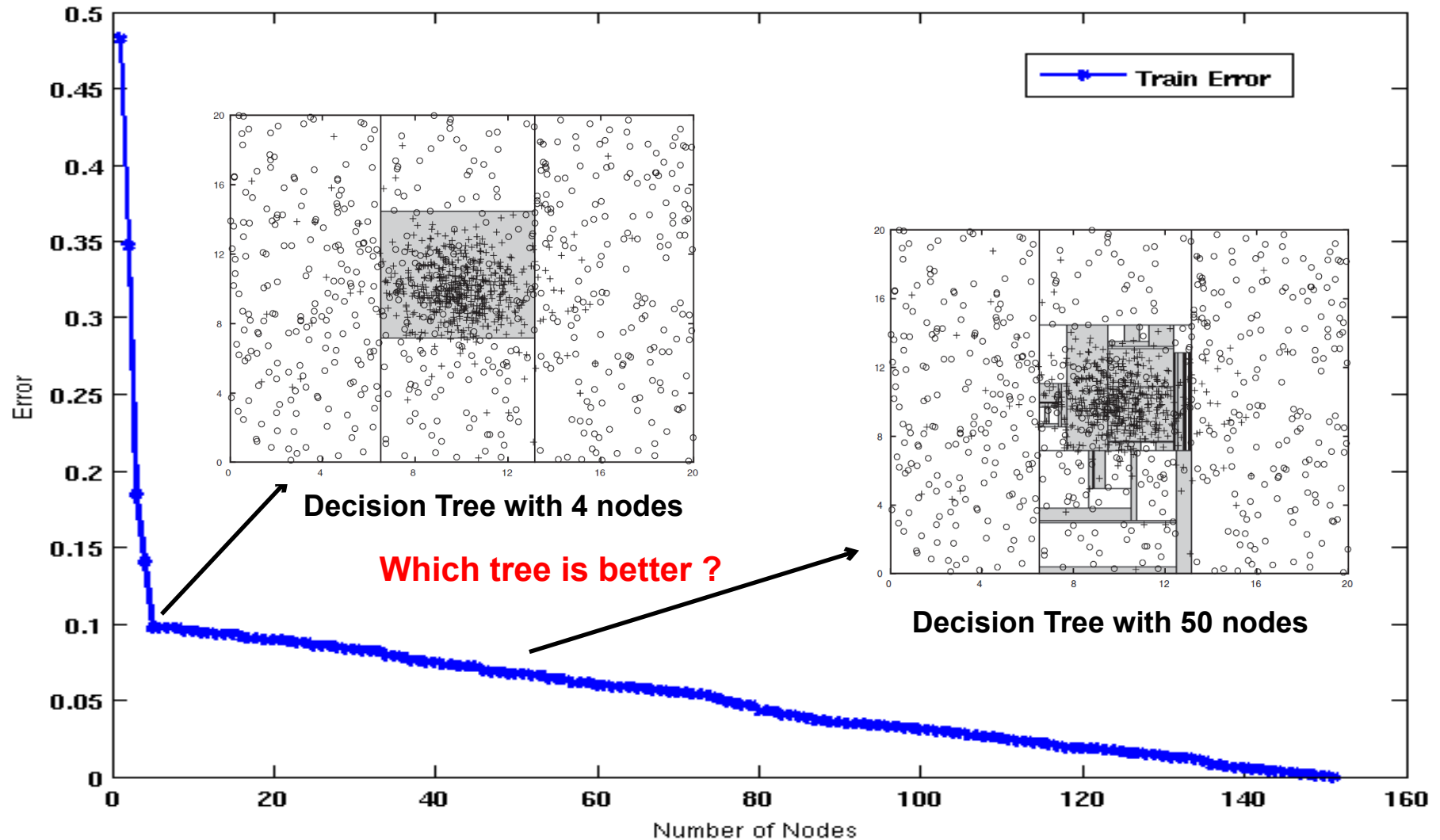
Decision Tree with 4 nodes



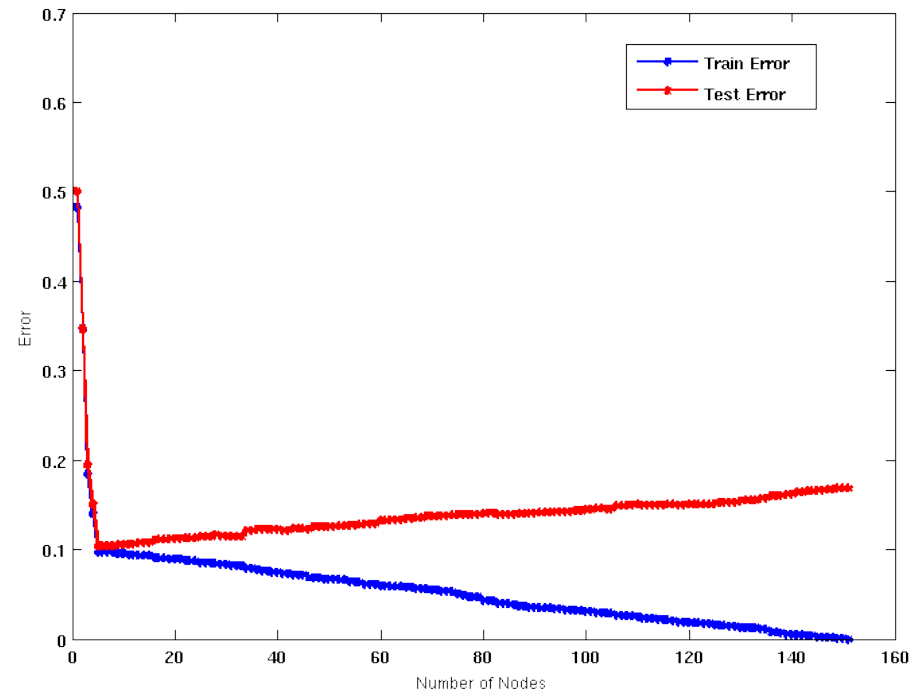
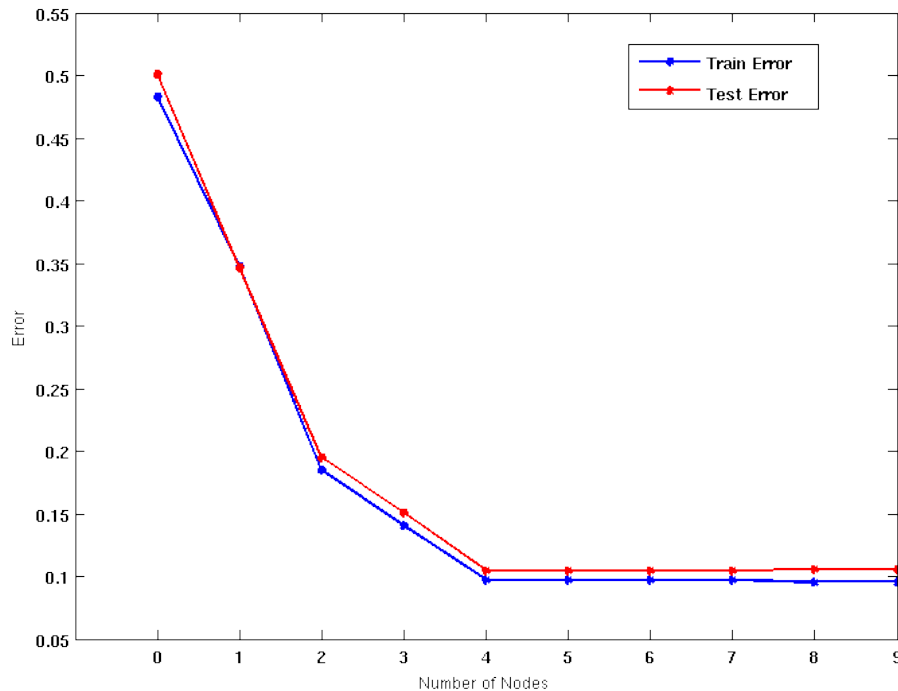
Decision Tree with 50 nodes



Which tree is better?



Model Underfitting and Overfitting

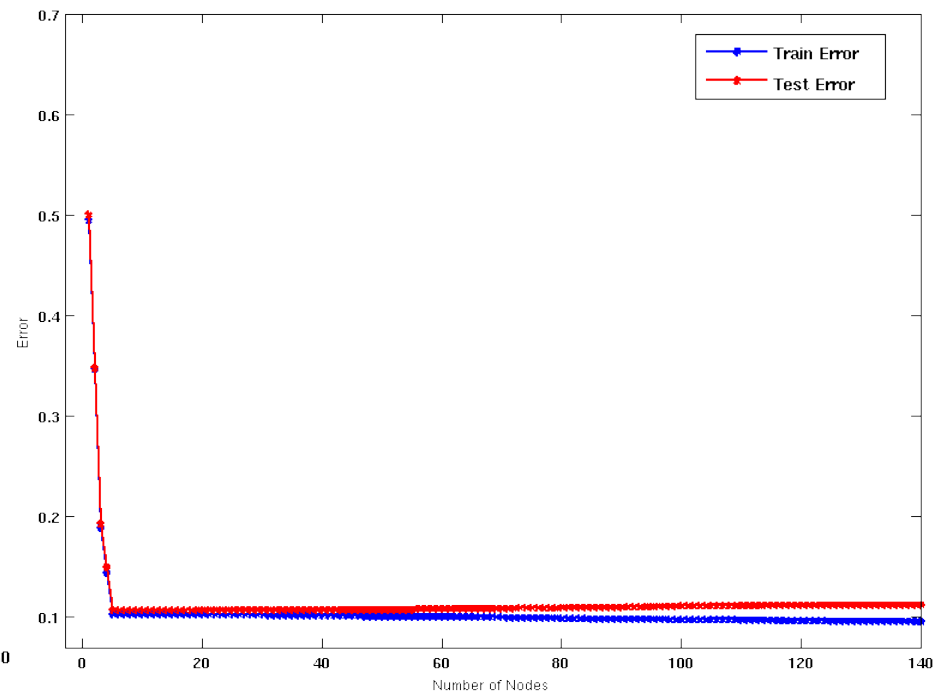
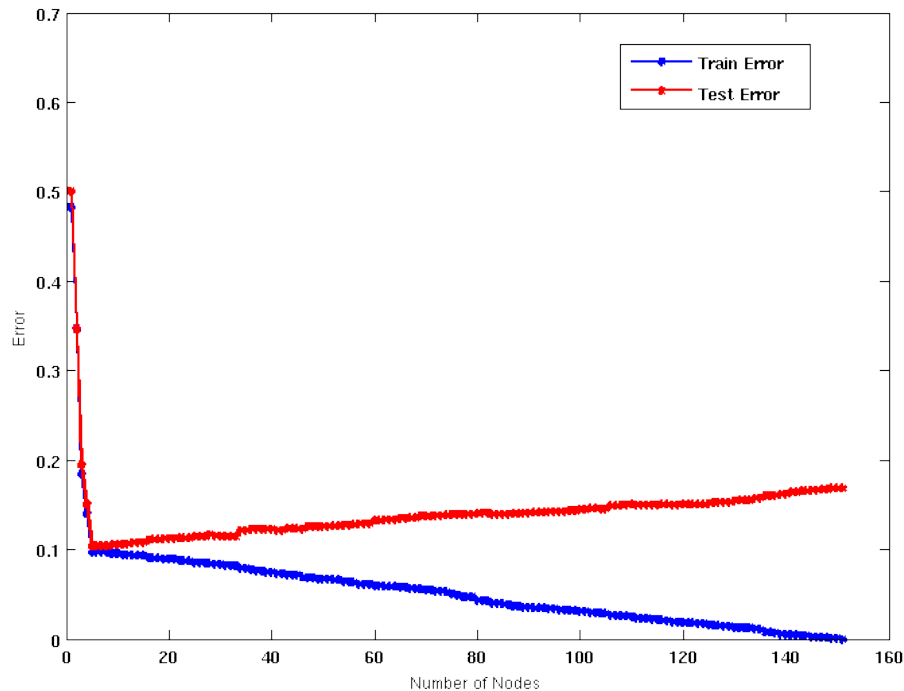


- As the model becomes more and more complex, test errors can start increasing even though training error may be decreasing

Underfitting: when model is too simple, both training and test errors are large

Overfitting: when model is too complex, training error is small but test error is large

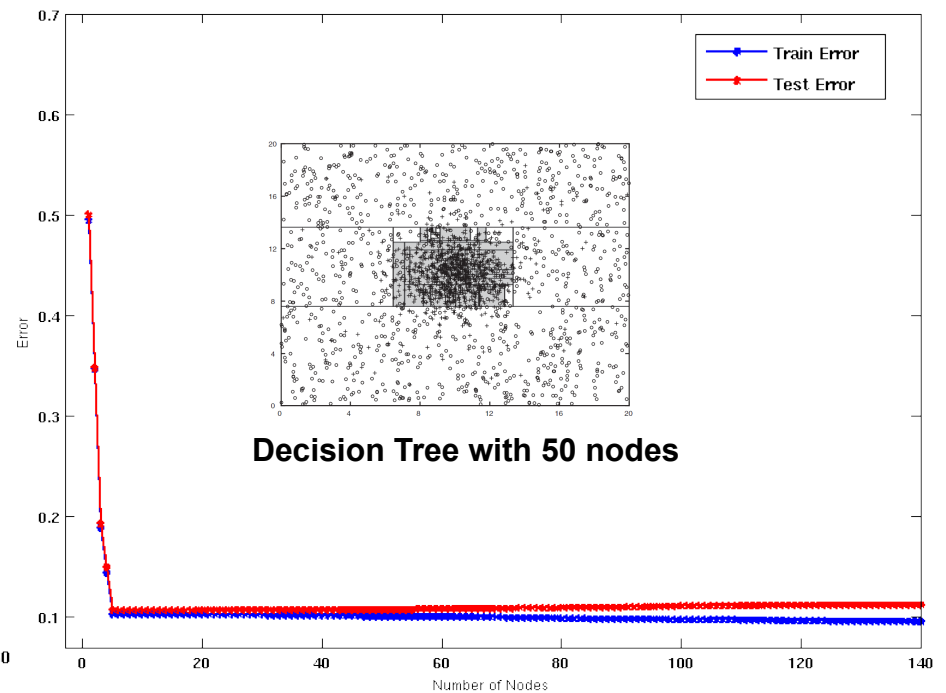
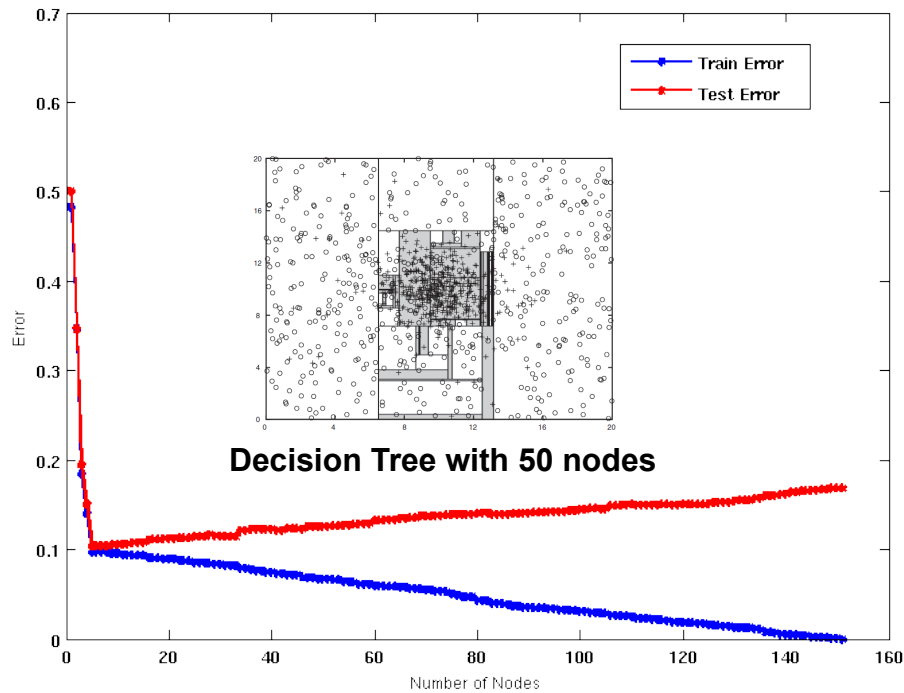
Model Overfitting – Impact of Training Data Size



Using twice the number of data instances

- Increasing the size of training data reduces the difference between training and testing errors at a given size of model

Model Overfitting – Impact of Training Data Size



Using twice the number of data instances

- Increasing the size of training data reduces the difference between training and testing errors at a given size of model

Reasons for Model Overfitting

- Not enough training data
- High model complexity
 - Multiple Comparison Procedure

Notes on Overfitting

- Overfitting results in decision trees that are more complex than necessary
- Training error does not provide a good estimate of how well the tree will perform on previously unseen records
- Need ways for estimating generalization errors

Model Selection

- Performed during model building
- Purpose is to ensure that model is not overly complex (to avoid overfitting)
- Need to estimate generalization error
 - Using Validation Set
 - Incorporating Model Complexity

Model Selection:

Using Validation Set

- Divide training data into two parts:
 - Training set:
 - ◆ use for model building
 - Validation set:
 - ◆ use for estimating generalization error
 - ◆ Note: validation set is not the same as test set
- Drawback:
 - Less data available for training

Model Selection:

Incorporating Model Complexity

- Rationale: Occam's Razor
 - Given two models of similar generalization errors, one should prefer the simpler model over the more complex model
 - A complex model has a greater chance of being fitted accidentally
 - Therefore, one should include model complexity when evaluating a model

$$\text{Gen. Error}(\text{Model}) = \text{Train. Error}(\text{Model}, \text{Train. Data}) + \alpha \times \text{Complexity}(\text{Model})$$

Model Selection for Decision Trees

- Pre-Pruning (Early Stopping Rule)

- Stop the algorithm before it becomes a fully-grown tree
- Typical stopping conditions for a node:
 - ◆ Stop if all instances belong to the same class
 - ◆ Stop if all the attribute values are the same
- More restrictive conditions:
 - ◆ Stop if number of instances is less than some user-specified threshold
 - ◆ Stop if class distribution of instances are independent of the available features (e.g., using χ^2 test)
 - ◆ Stop if expanding the current node does not improve impurity measures (e.g., Gini or information gain).
 - ◆ Stop if estimated generalization error falls below certain threshold

Model Selection for Decision Trees

- Post-pruning

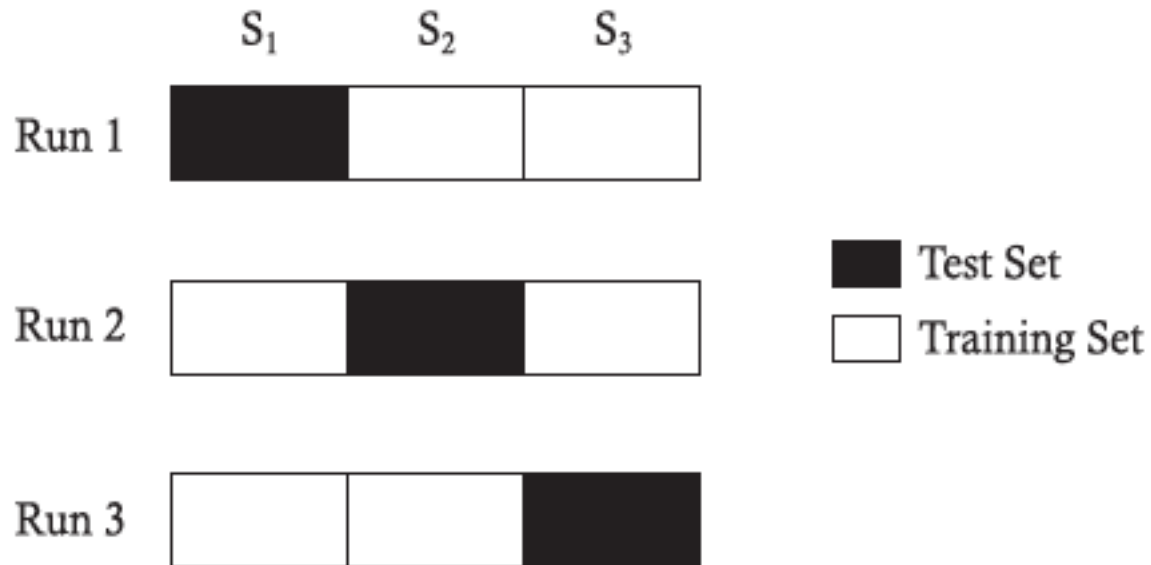
- Grow decision tree to its entirety
- Subtree replacement
 - ◆ Trim the nodes of the decision tree in a bottom-up fashion
 - ◆ If generalization error improves after trimming, replace sub-tree by a leaf node
 - ◆ Class label of leaf node is determined from majority class of instances in the sub-tree

Model Evaluation

- Purpose:
 - To estimate performance of classifier on previously unseen data (test set)
- Holdout
 - Reserve $k\%$ for training and $(100-k)\%$ for testing
 - Random subsampling: repeated holdout
- Cross validation
 - Partition data into k disjoint subsets
 - k -fold: train on $k-1$ partitions, test on the remaining one
 - Leave-one-out: $k=n$

Cross-validation Example

- 3-fold cross-validation



Variations on Cross-validation

- Repeated cross-validation
 - Perform cross-validation a number of times
 - Gives an estimate of the variance of the generalization error
- Stratified cross-validation
 - Guarantee the same percentage of class labels in training and test
 - Important when classes are imbalanced and the sample is small
- Use nested cross-validation approach for model selection and evaluation



Model Evaluation

Class Imbalance Problem

- Lots of classification problems where the classes are skewed (more records from one class than another)
 - Credit card fraud
 - Intrusion detection
 - Defective products in manufacturing assembly line
 - COVID-19 test results on a random sample
- **Key Challenge:**
 - Evaluation measures such as accuracy are not well-suited for imbalanced class

Confusion Matrix

- Confusion Matrix:

ACTUAL CLASS	PREDICTED CLASS	
	Class=Yes	Class=No
Class=Yes	a	b
Class=No	c	d

a: TP (true positive)

b: FN (false negative)

c: FP (false positive)

d: TN (true negative)

Accuracy

ACTUAL CLASS	PREDICTED CLASS	
	Class=Yes	Class=No
Class=Yes	a (TP)	b (FN)
	c (FP)	d (TN)

- Most widely-used metric:

$$\text{Accuracy} = \frac{a + d}{a + b + c + d} = \frac{TP + TN}{TP + TN + FP + FN}$$

Problem with Accuracy

- Consider a 2-class problem
 - Number of Class NO examples = 990
 - Number of Class YES examples = 10
- If a model predicts everything to be class NO, accuracy is $990/1000 = 99\%$
 - This is misleading because this trivial model does not detect any class YES example
 - Detecting the rare class is usually more interesting (e.g., frauds, intrusions, defects, etc)

	PREDICTED CLASS		
ACTUAL CLASS		Class=Yes	Class=No
	Class=Yes	0	10
	Class=No	0	990

Which model is better?

A

ACTUAL	PREDICTED		
		Class=Yes	Class=No
	Class=Yes	0	10
	Class=No	0	990

Accuracy: 99%

B

ACTUAL	PREDICTED		
		Class=Yes	Class=No
	Class=Yes	10	0
	Class=No	100	890

Accuracy: 90%

Alternative Measures

	PREDICTED CLASS		
ACTUAL CLASS		Class=Yes	Class=No
	Class=Yes	a	b
	Class=No	c	d

$$\text{Precision (p)} = \frac{a}{a + c}$$

$$\text{Recall (r)} = \frac{a}{a + b}$$

$$\text{F - measure (F)} = \frac{2rp}{r + p} = \frac{2a}{2a + b + c}$$

Alternative Measures

	PREDICTED CLASS		
		Class=Yes	Class=No
	Class=Yes	10	0
	Class=No	10	980

$$\text{Precision (p)} = \frac{10}{10+10} = 0.5$$

$$\text{Recall (r)} = \frac{10}{10+0} = 1$$

$$\text{F - measure (F)} = \frac{2 * 1 * 0.5}{1 + 0.5} = 0.62$$

$$\text{Accuracy} = \frac{990}{1000} = 0.99$$

Alternative Measures

ACTUAL CLASS	PREDICTED CLASS		
		Class=Yes	Class=No
	Class=Yes	10	0
	Class=No	10	980

$$\text{Precision (p)} = \frac{10}{10+10} = 0.5$$

$$\text{Recall (r)} = \frac{10}{10+0} = 1$$

$$\text{F - measure (F)} = \frac{2 * 1 * 0.5}{1 + 0.5} = 0.62$$

$$\text{Accuracy} = \frac{990}{1000} = 0.99$$

ACTUAL CLASS	PREDICTED CLASS		
		Class=Yes	Class=No
	Class=Yes	1	9
	Class=No	0	990

$$\text{Precision (p)} = \frac{1}{1+0} = 1$$

$$\text{Recall (r)} = \frac{1}{1+9} = 0.1$$

$$\text{F - measure (F)} = \frac{2 * 0.1 * 1}{1 + 0.1} = 0.18$$

$$\text{Accuracy} = \frac{991}{1000} = 0.991$$